Linux'da Oyunlar



Onur Küçük <onur@delipenguen.net>

Oyun ?

- Herkes oyun oynar
- Bilgisayar alırken oyun için fazla para verilir
- Çocukların bilgisayarla oyun aracılığıyla tanışı



Tarihçe

- 1995 – Klasik X oyunları
- 1997 – Kaliteli oyunlar için çalışmalar
- 1999 – Ticari destek (Quake serisi, Unreal)
- 2004 – UT2004, Neverwinter Nights, Cube



Neler Gerekli

- Amaca hitap eden donanım
- Ayarları yapılmış kaliteli bir işletim sistemi
- Gerekli yazılımlar
- Oyun



Donanım Desteği (Ekran Kartı)

- Nvidia
- Ati
- Matrox
- Intel
- DRI (http://dri.sourceforge.net)



Donanım Desteği (Ses)

- OSS
- ALSA
- AC97
- Creative, Yamaha, ESS ...



Donanım Desteği (Joystick/Gamepad)

- Çekirdek tarafından desteklenen herhangi bir Jostick/Gamepad
- Creative Labs Blaster Cobra
- MS Sidewinder (forcefeedback desteği ile)
- Logitech Wingman Warrior
- Multisystem Sega, NES, PS



Performans

- Hangi dağıtım ?
- Eksik/yanlış sürüm kütüphaneler
- Gereksiz yükleri azaltmak (Masaüstü ortamı, ses sunucusu, sistem servisleri)
- Doğru arabirim (Xmame/GLmame, wine/winex)



Oyun Seçerken

- Linux sürümü
- Linux çalıştırılabilir dosya + Başka OS CD'si
- Emulator + Başka OS/mimari CD'si



Neden Linux'da Oyun ?

- Kararlılık
- Performans
- Çok kullanıcılı sistem
- Çoklu platform desteği
- DirectX vs. OpenGL



Neden Açık Kod Oyunlar

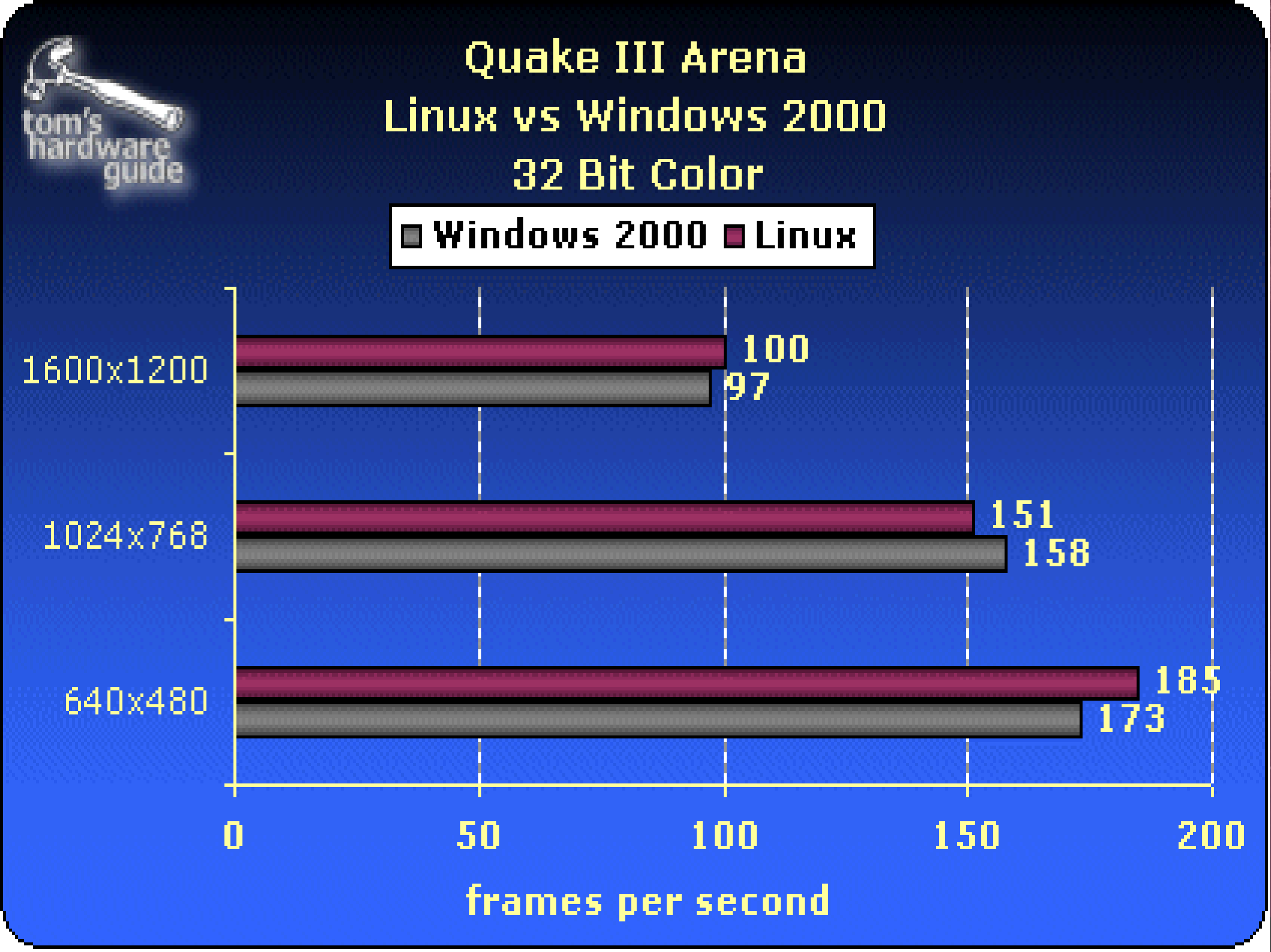
- Hızlı güncelleme, hata düzeltme
- Multiplatform
- Sorun oluşursa çabuk müdahale
- Lisans ücreti

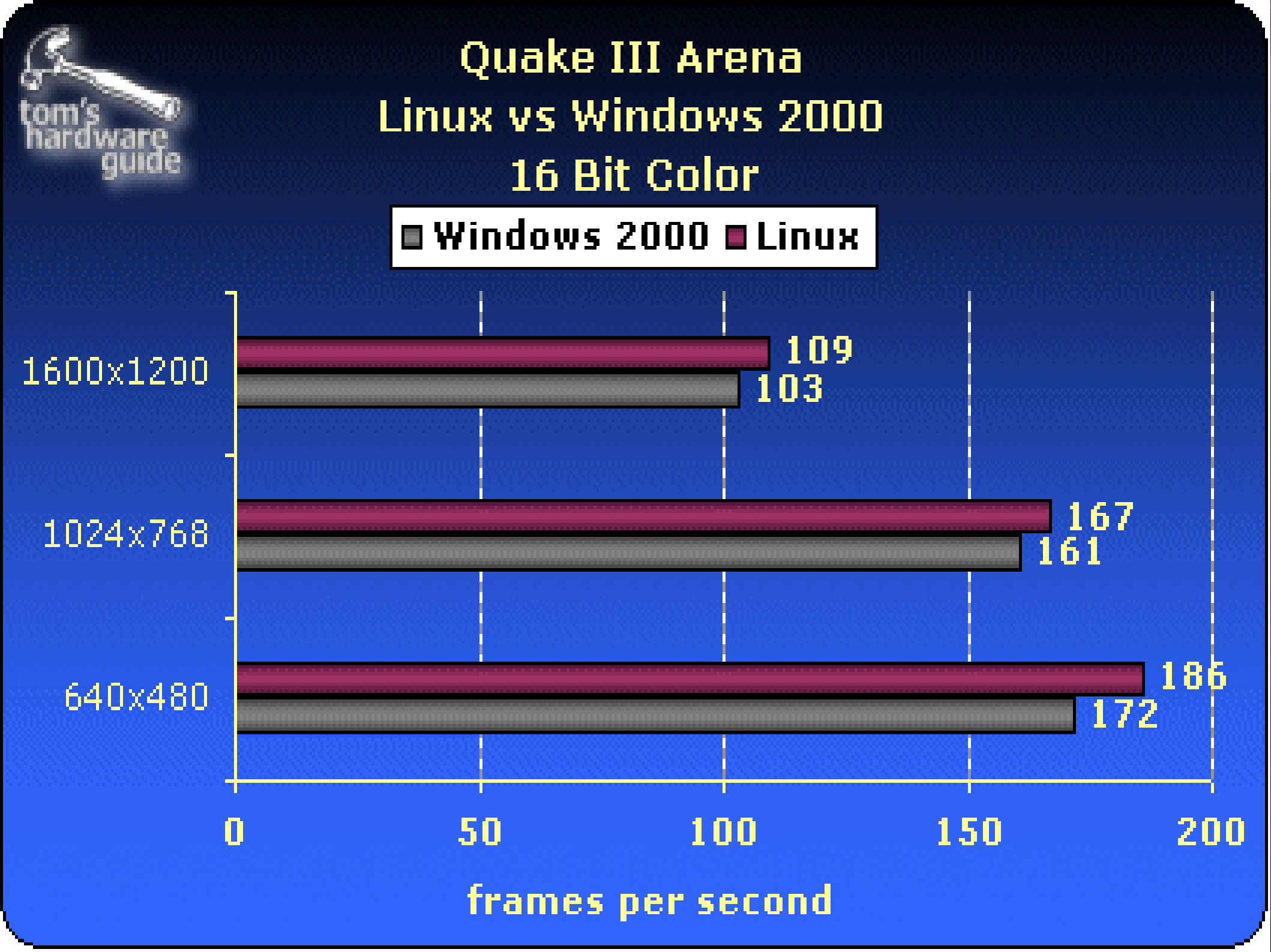


Örnek Test (tomshardware)

- AMD Athlon 1800+ (1.53 Ghz)
- 256MB DDR SD-RAM
- GeForce3 Ti 200
- 9GB SCSI 10000 RPM Seagate HDD
- MDK 8.2 vs. W2K
- Quake 3 v1.11
- Yüksek Kalite grafik ayarı
- demo001




tom's hardware guide
Quake III Arena
Linux vs Windows 2000
32 Bit Color
Windows 2000
Linux
1600x1200
100
97
1024x768
151
158
640x480
185
173
0
50
100
150
200
frames per second

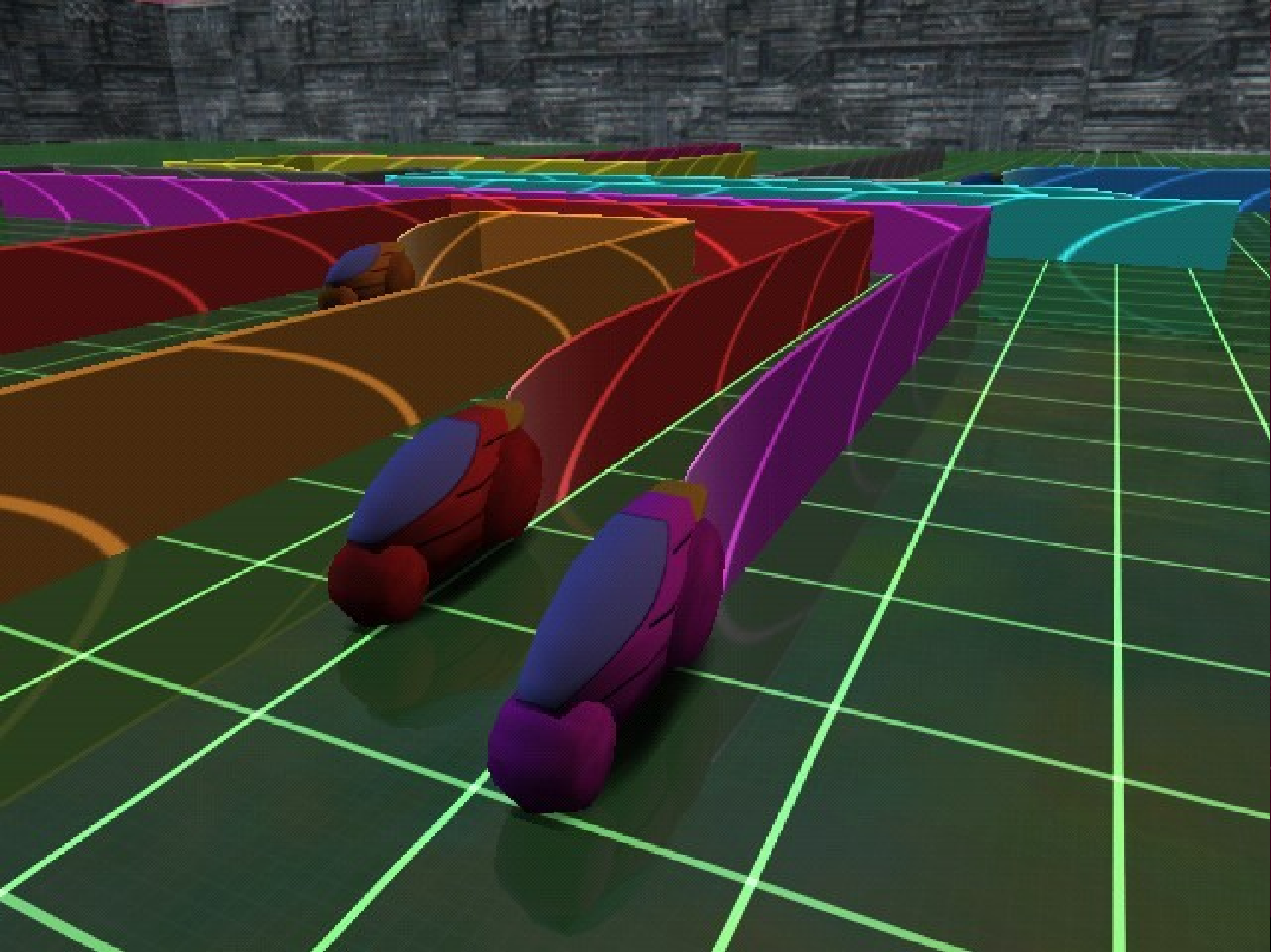

tom's hardware guide
Quake III Arena
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16 Bit Color
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Linux
1600x1200
109
103
1024x768
167
161
640x480
186
172
0
50
100
150
200
frames per second

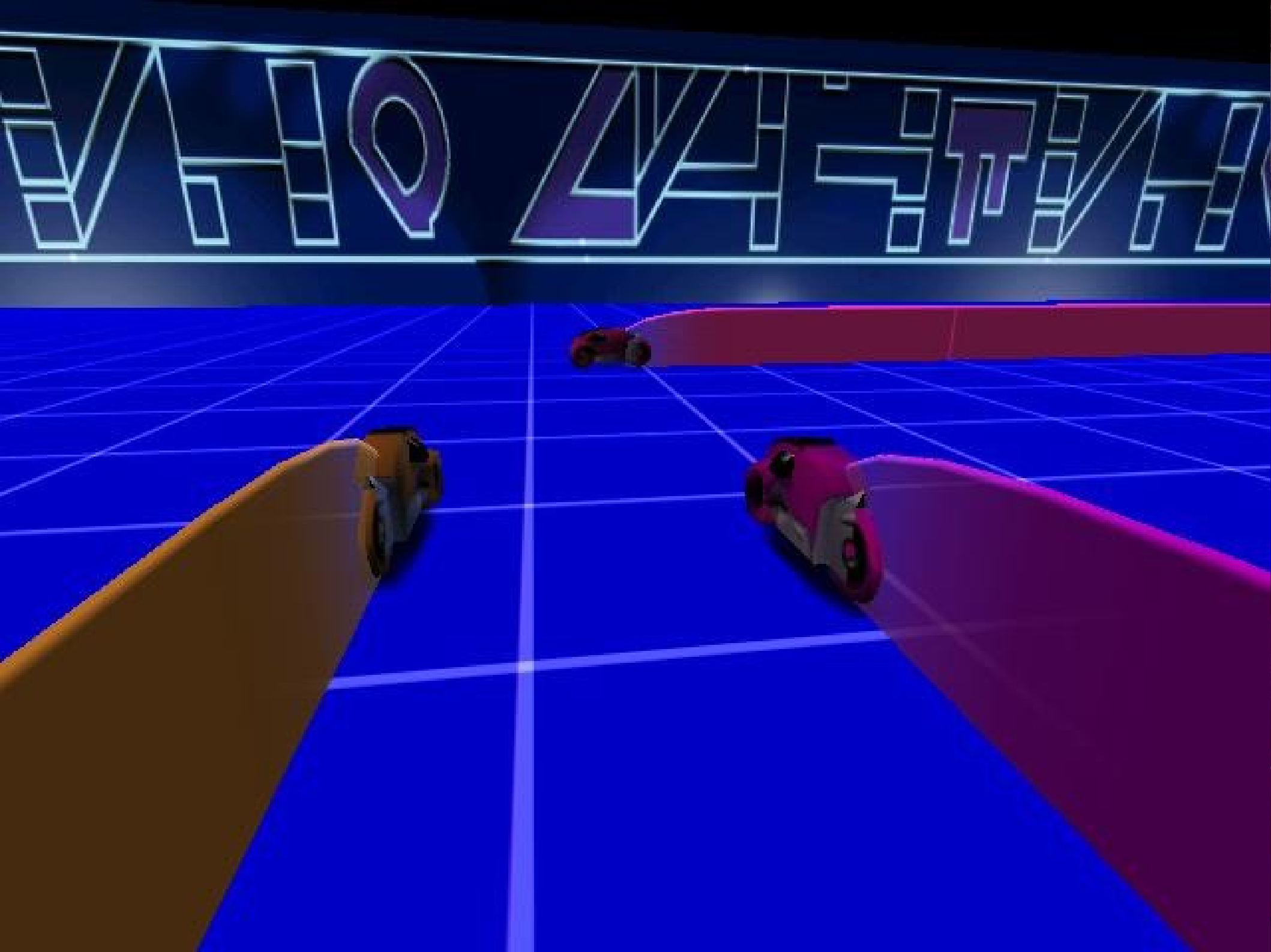
Açık Kaynak Kodlu Oyunlar ?



Armagetron







BZ Flag



Driving with jdoe

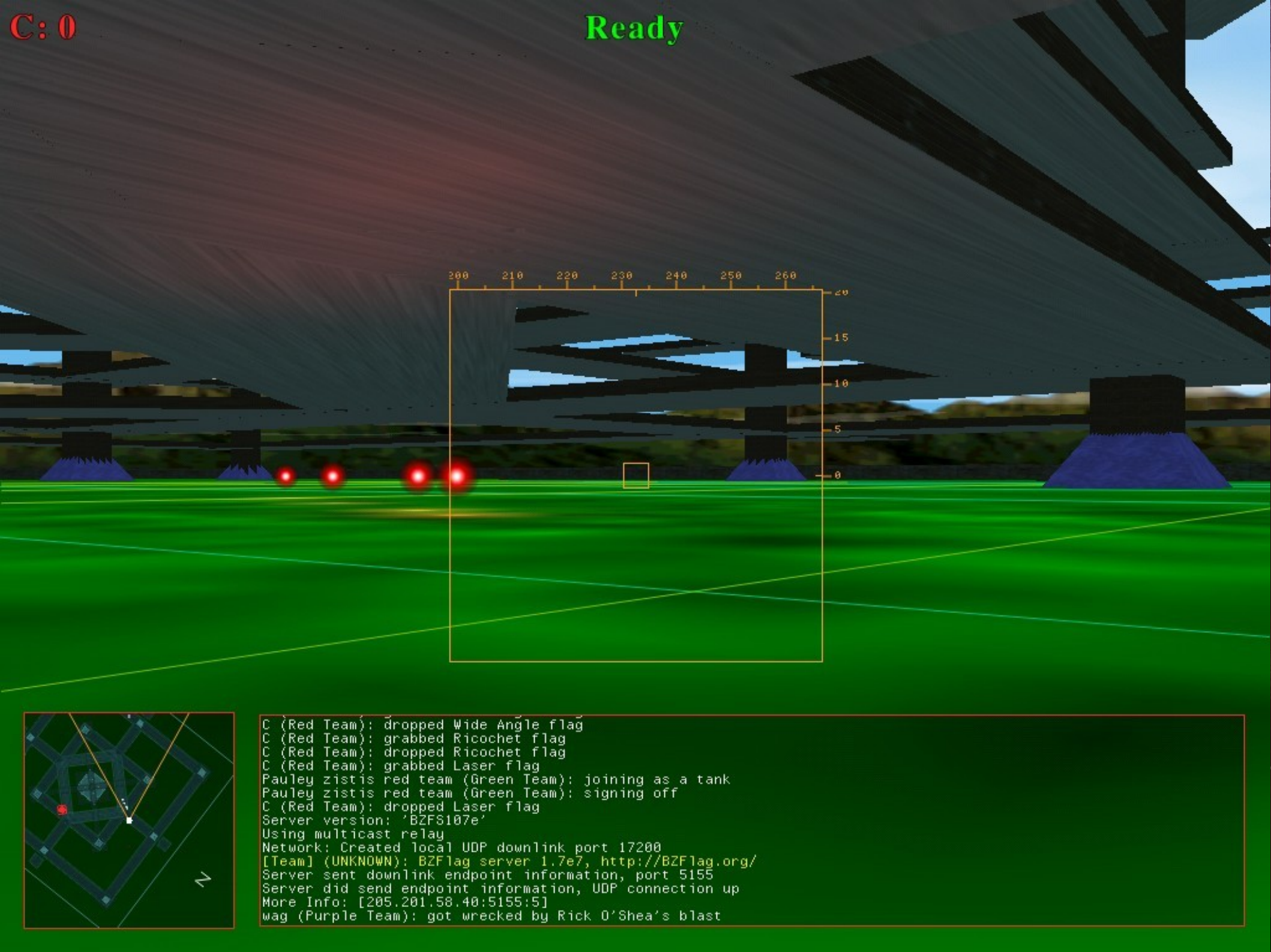


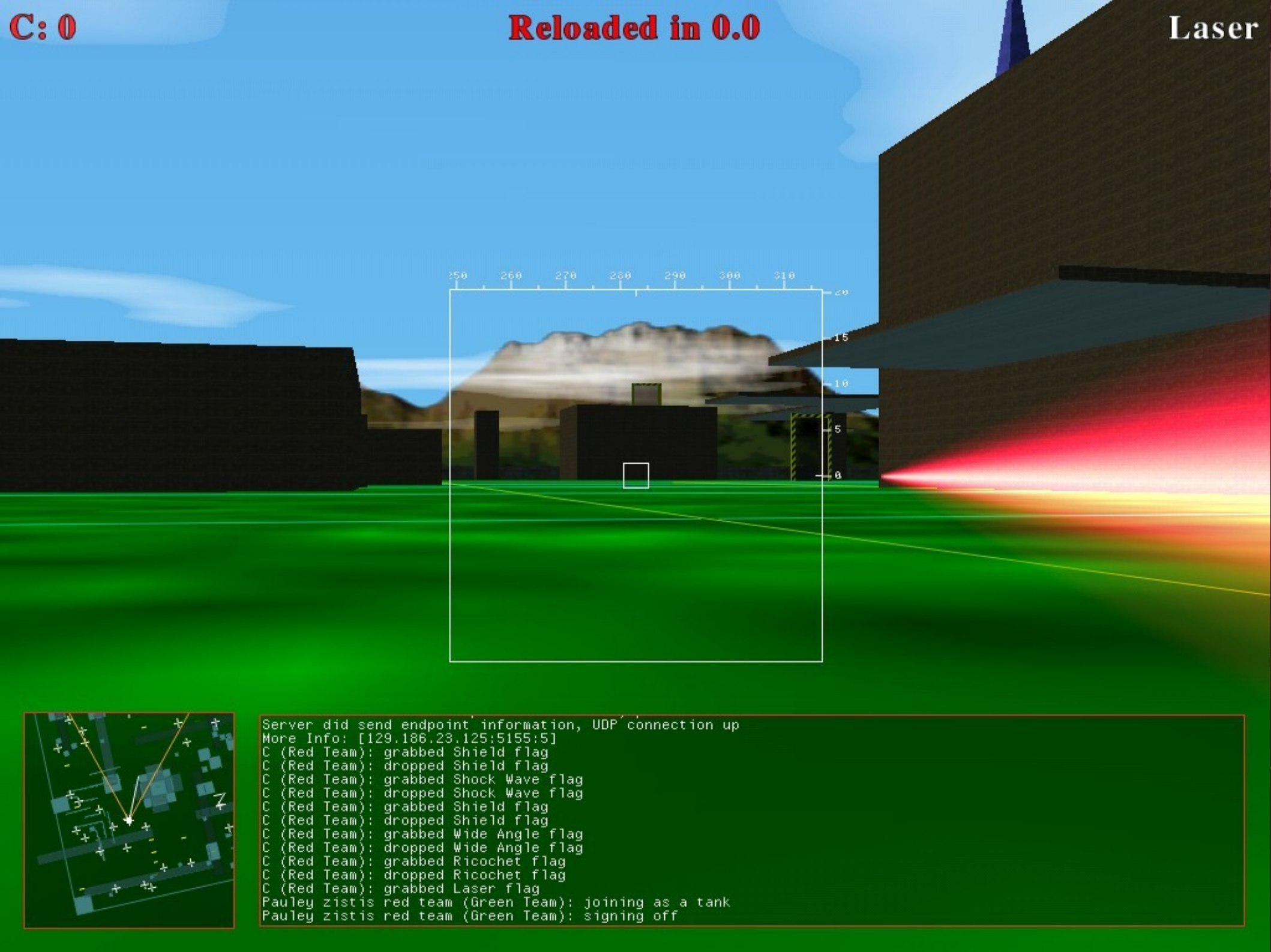
RozBoz (Green Team): got wrecked by jdoe's blast
Chris (Red Team): grabbed Laser flag
Kid Icuras (Purple Team): got wrecked by hate me!'s blast
jdoe (Purple Team): got wrecked by Blaster's blast
Blaster (Rogue): grabbed Machine Gun flag
Blaster (Rogue): dropped Machine Gun flag
mwa (Green Team): grabbed Steamroller flag
Chris (Red Team): dropped Laser flag
Chris (Red Team): was destroyed by Jack's guided missile
jdoe (Purple Team): grabbed Tiny flag
Kid Icuras (Purple Team): got wrecked by [dmp]'s blast
mwa (Green Team): dropped Steamroller flag
mwa (Green Team): was destroyed by Jack's guided missile
[dmp] (Rogue): got wrecked by RozBoz's blast
RozBoz (Green Team): got wrecked by jdoe's blast

Driving with Chris




C: -2
Reloaded in 2.8
Locked on Slug (Red Team)
Guided Missile
350
0
10
20
30
40
50
20
15
10
5
0
N
C (Purple Team): grabbed Guided Missile flag
Slug (Red Team): grabbed Rapid Fire flag
terebi ga yakikorosu (Green Team): dropped Tiny flag
Locked on terebi ga yakikorosu (Green Team)
terebi ga yakikorosu (Green Team): got wrecked by Aquitaine's blast
Slug (Red Team): dropped Rapid Fire flag
Slug (Red Team): got wrecked by teammate's blast
Aquitaine (Purple Team): was destroyed by teammate C's guided missile
Locked on Slug (Red Team)
terebi ga yakikorosu (Green Team): grabbed Machine Gun flag
Dragon (Red Team): got wrecked by teammate's blast
teammate (Blue Team): dropped Stealth flag
teammate (Blue Team): got wrecked by mofo's blast
tommy ramone (Blue Team): dropped Shock Wave flag
tommy ramone (Blue Team): destroyed by mofo


C: 0
Ready
200
210
220
230
240
250
260
20
15
10
5
0
C (Red Team): dropped Wide Angle flag
C (Red Team): grabbed Ricochet flag
C (Red Team): dropped Ricochet flag
C (Red Team): grabbed Laser flag
Pauley zistis red team (Green Team): joining as a tank
Pauley zistis red team (Green Team): signing off
C (Red Team): dropped Laser flag
Server version: 'BZFS107e'
Using multicast relay
Network: Created local UDP downlink port 17200
[Team] (UNKNOWN): BZFlag server 1.7e7, http://BZFlag.org/
Server sent downlink endpoint information, port 5155
Server did send endpoint information, UDP connection up
More Info: [205.201.58.40:5155:5]
wag (Purple Team): got wrecked by Rick O'Shea's blast


C: 0
Reloaded in 0.0
Laser
250
260
270
280
290
300
310
20
15
10
5
0
Server did send endpoint information, UDP connection up
More Info: [129.186.23.125:5155:5]
C (Red Team): grabbed Shield flag
C (Red Team): dropped Shield flag
C (Red Team): grabbed Shock Wave flag
C (Red Team): dropped Shock Wave flag
C (Red Team): grabbed Shield flag
C (Red Team): dropped Shield flag
C (Red Team): grabbed Wide Angle flag
C (Red Team): dropped Wide Angle flag
C (Red Team): grabbed Ricochet flag
C (Red Team): dropped Ricochet flag
C (Red Team): grabbed Laser flag
Pauley zistis red team (Green Team): joining as a tank
Pauley zistis red team (Green Team): signing off

Chromium B.S.U.




50.0
0016250

Egoboo




Get away from me!
White Beard killed a Lumpkin!
Twugg
$ 0
20
White
$ 0
8
GROG

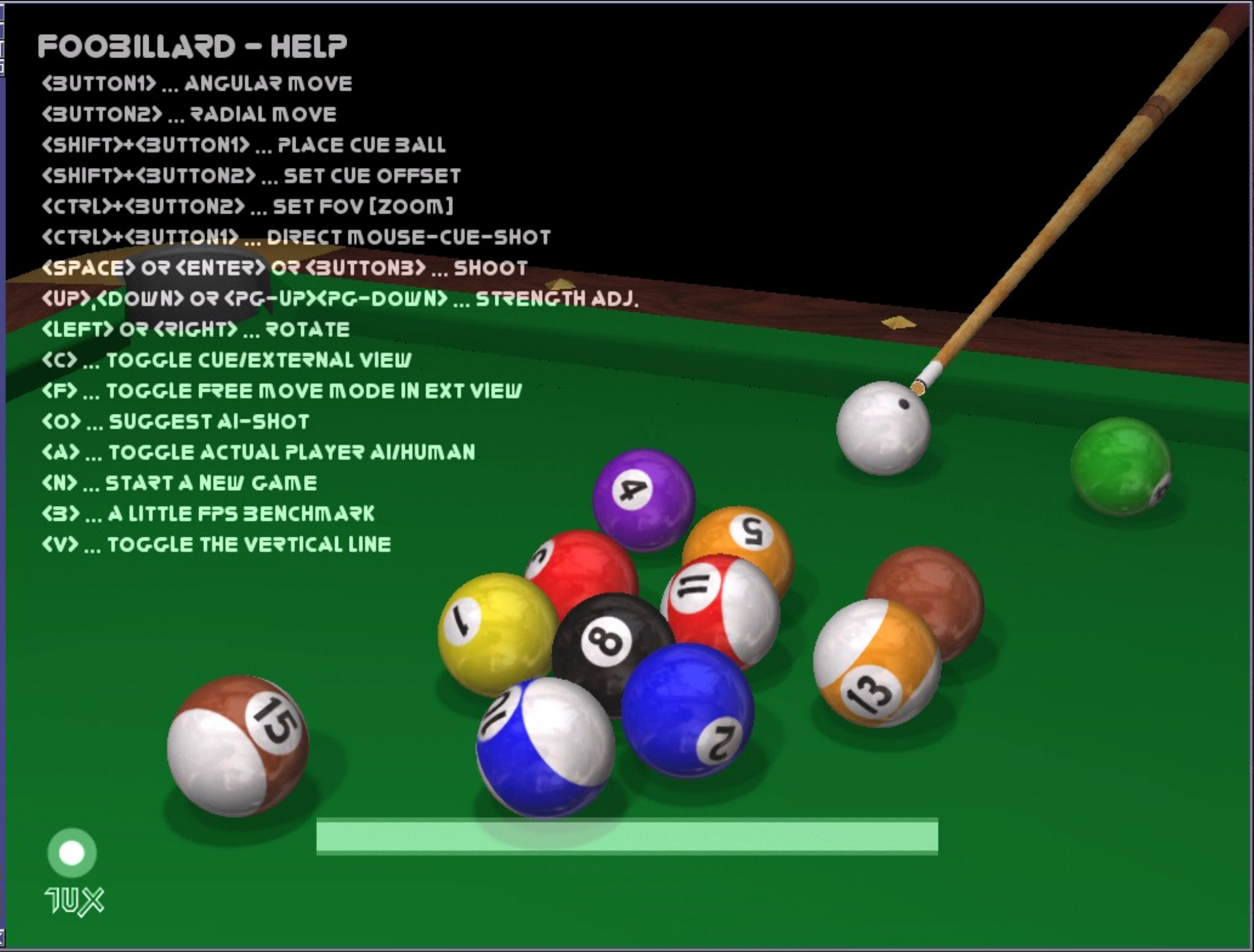

025 FPS
Plaggy
$ 14
Flambi
$ 58


029 FPS
Zambi
$ 106


018 FPS
Plaggy
$ 14

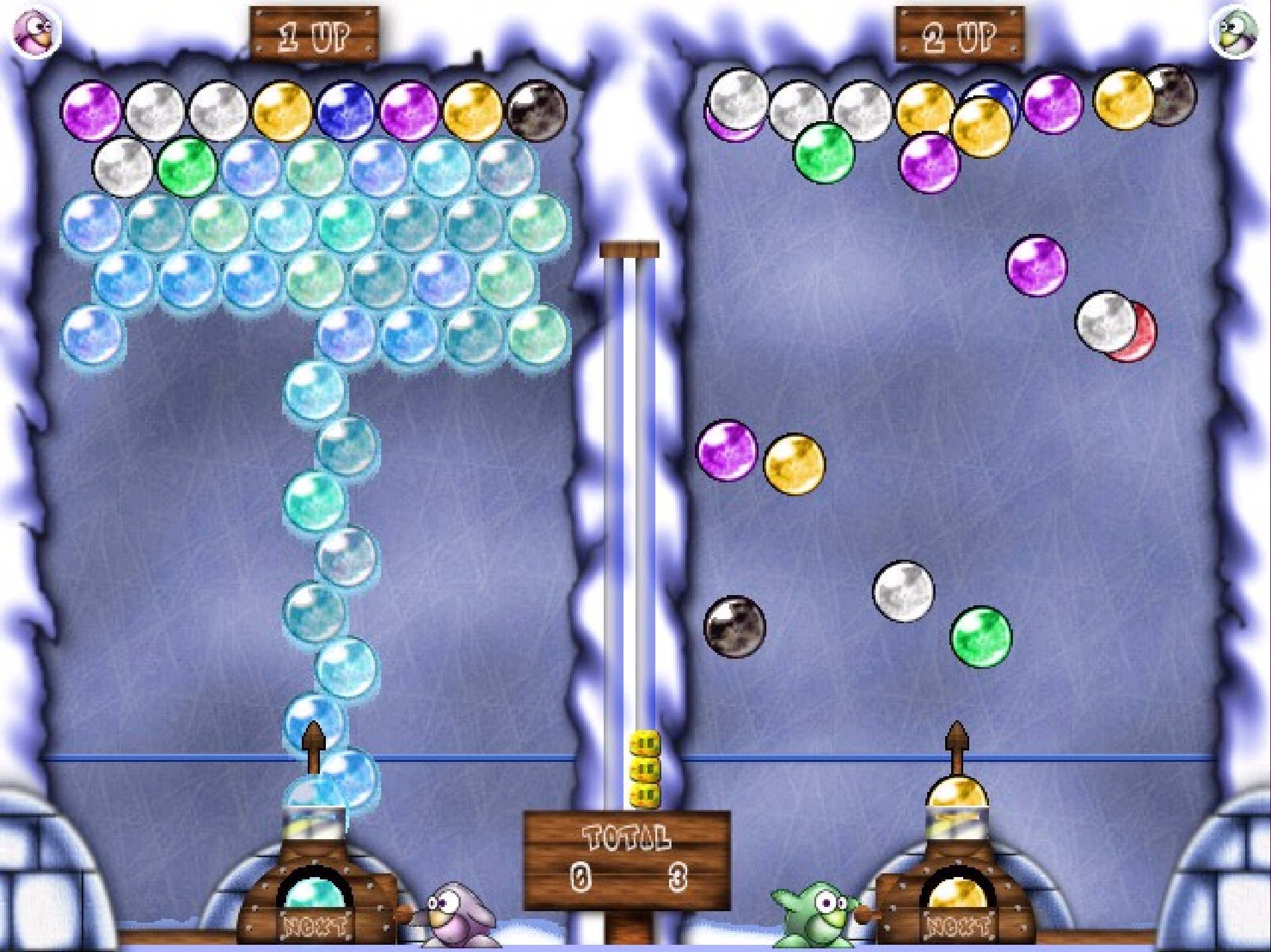
FooBilliard




FOOBILLARD - HELP
<BUTTON1> ... ANGULAR MOVE
<BUTTON2> ... RADIAL MOVE
<SHIFT>+<BUTTON1> ... PLACE CUE BALL
<SHIFT>+<BUTTON2> ... SET CUE OFFSET
<CTRL>+<BUTTON2> ... SET FOV [ZOOM]
<CTRL>+<BUTTON1> ... DIRECT MOUSE-CUE-SHOT
<SPACE> OR <ENTER> OR <BUTTON3> ... SHOOT
<UP>,<DOWN> OR <PG-UP><PG-DOWN> ... STRENGTH ADJ.
<LEFT> OR <RIGHT> ... ROTATE
<C> ... TOGGLE CUE/EXTERNAL VIEW
<F> ... TOGGLE FREE MOVE MODE IN EXT VIEW
<O> ... SUGGEST AI-SHOT
<A> ... TOGGLE ACTUAL PLAYER AI/HUMAN
<N> ... START A NEW GAME
<B> ... A LITTLE FPS BENCHMARK
<V> ... TOGGLE THE VERTICAL LINE
TUX

Frozen Bubble




1 UP
2 UP
TOTAL
0 3

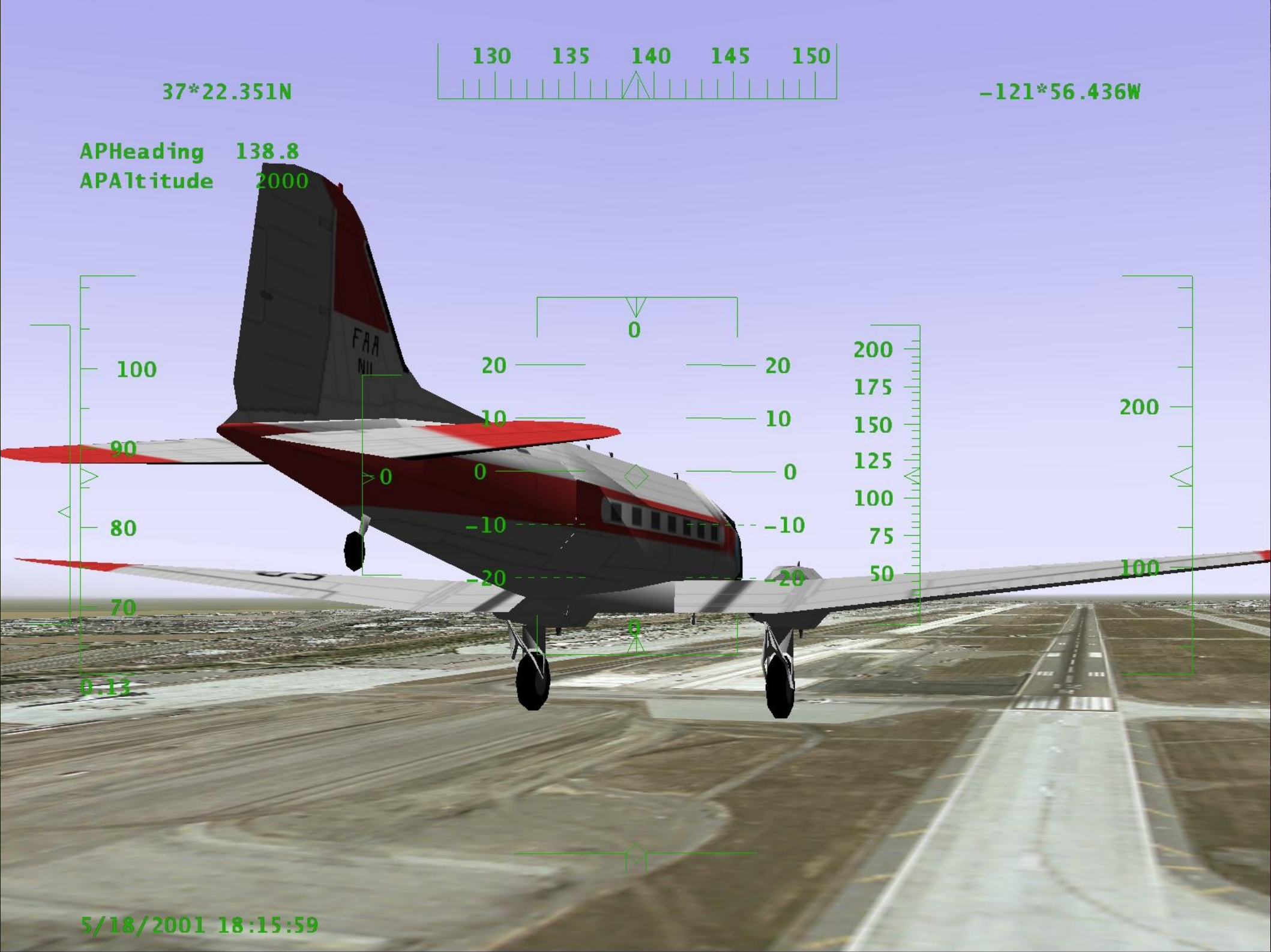
Flight Gear





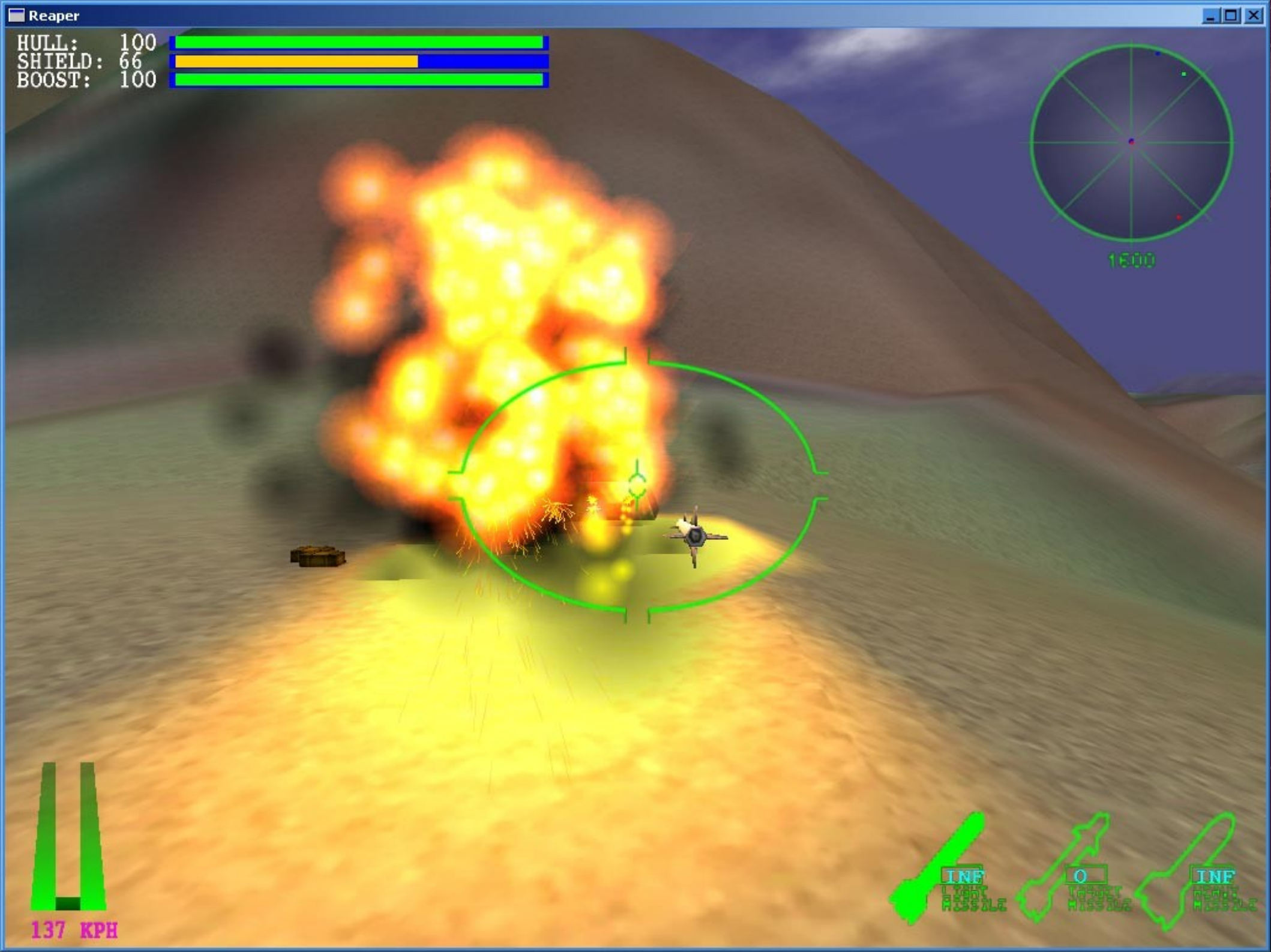

12
21


WM930
ROYAL NAVY
611


130 135 140 145 150
37*22.351N
-121*56.436W
APHeading 138.8
APAltitude 2000
100
90
80
70
0.13
0
20
10
0
-10
-20
0
20
10
0
-10
-20
0
200
175
150
125
100
75
50
200
100
5/18/2001 18:15:59

Reaper




Reaper
HULL: 100
SHIELD: 66
BOOST: 100
1600
INF
0
INF
137 KPH



Pingus



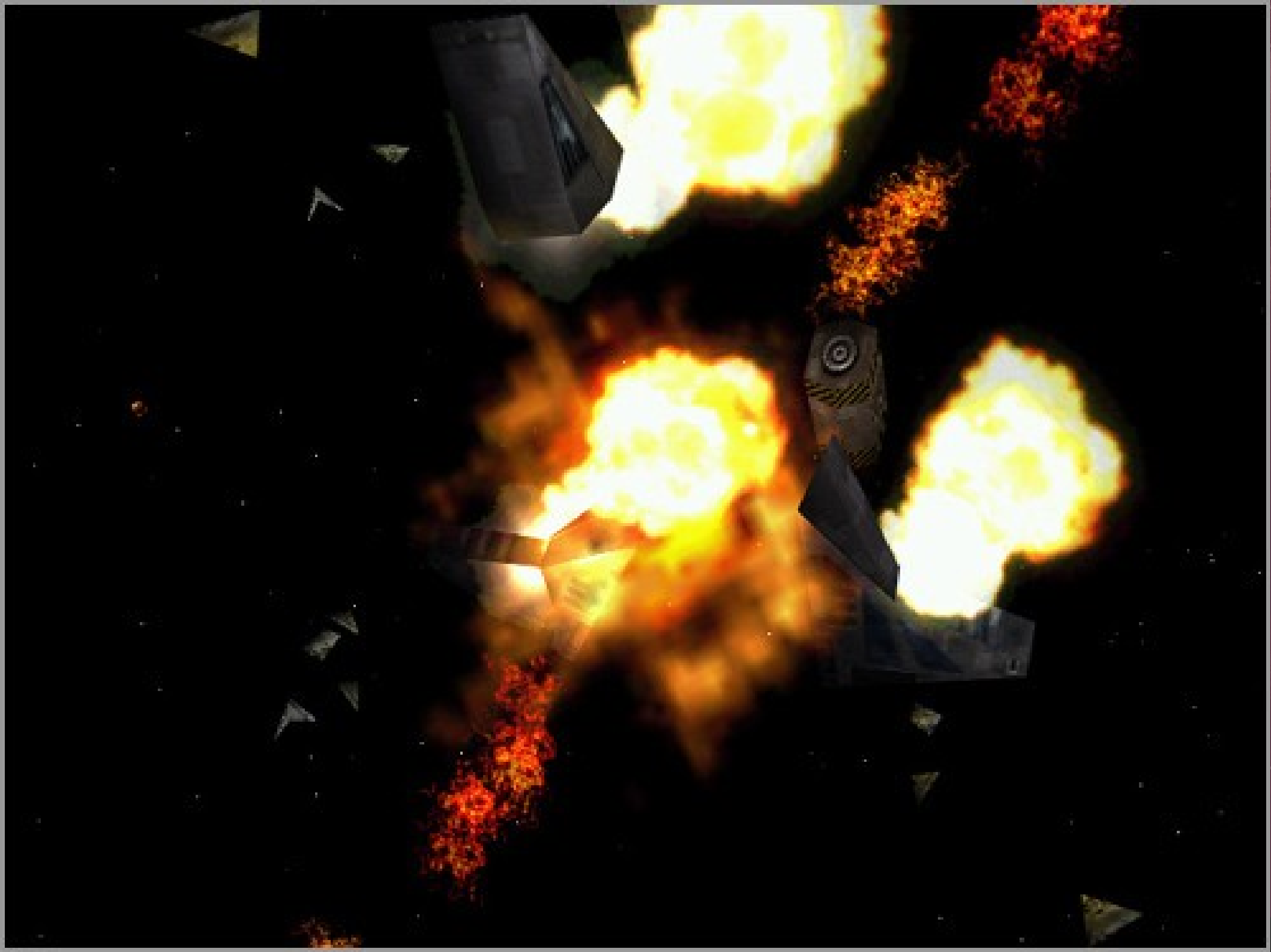

Released: 7/25
Out: 5
Saved: 2/5
00
73

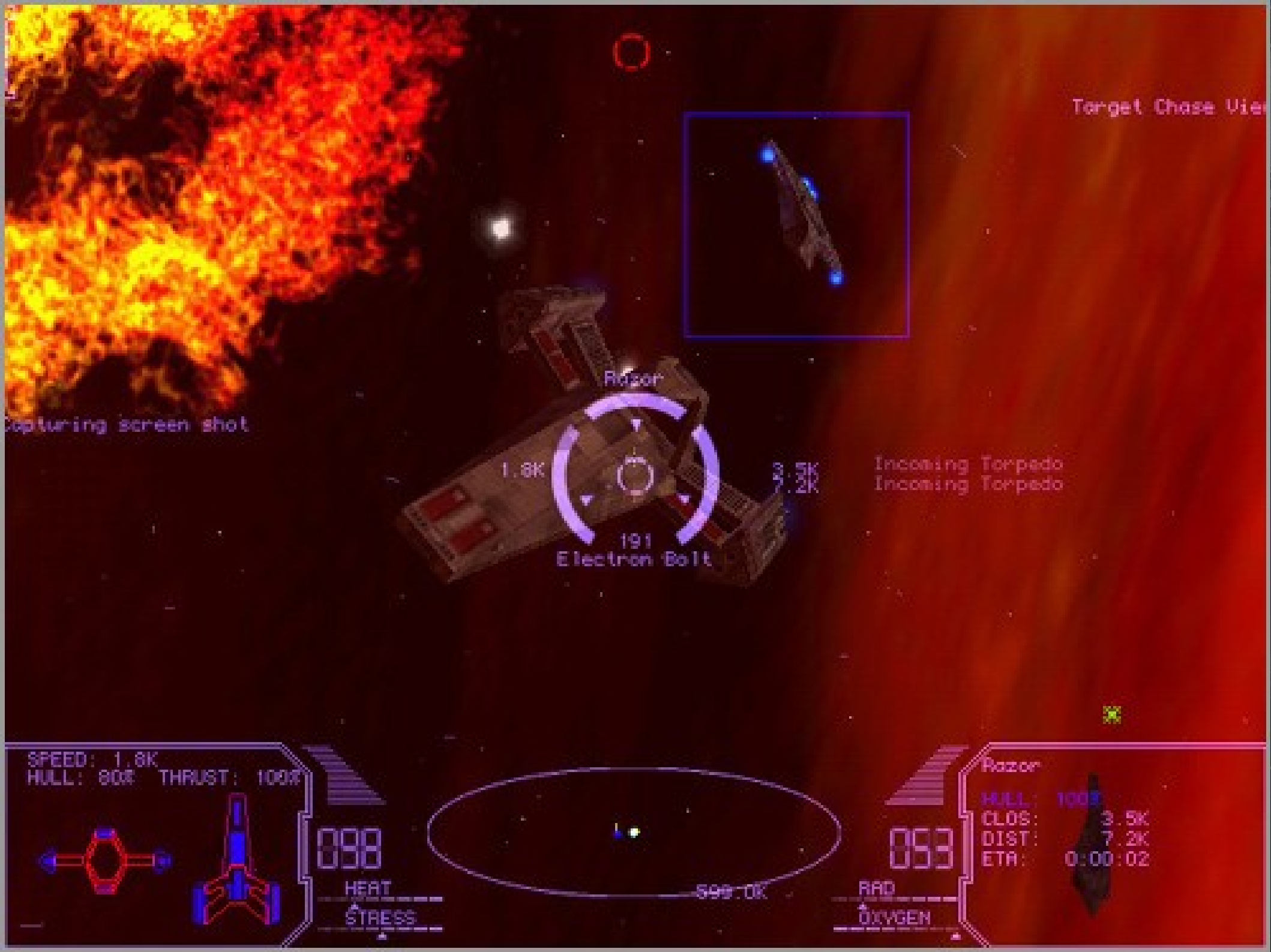

Released: 10/10
Out: 7
Saved: 3/5
2:56
Walker Id: 8
EXIT

Terminus




Target Chase Vie
BeltMarsGate
Targeting nearest object
5.4K
-5.1K
28.5K
1000
Proton Particle Bolt
SPEED: 5.4K
HULL: 100% THRUST: 100%
BeltMarsGate
CLOS: -5.1K
DIST: 28.5K
ETA: 0:00:00
HEAT
STRESS
RAD
OXYGEN




Target Chase Vie
Capturing screen shot
Razor
1.8K
3.5K
7.2K
191
Electron Bolt
Incoming Torpedo
Incoming Torpedo
SPEED: 1.8K
HULL: 80% THRUST: 100%
098
HEAT
STRESS
053
RAD
OXYGEN
Razor
HULL: 100%
CLOS: 3.5K
DIST: 7.2K
ETA: 0:00:02



Tuxpuck




Continue
Surrender

OpenMortal




14 fps
28

Vegastrike




Fitchburg:Fitchburg:unknown: 1.14
0.00 Welcome to Vegastrike
50
G: Laser,Laser
0
0
Fitchburg
Dis 48119.6328


Fitchburg:Fitchburg:unknown: 1.14
0.00 Welcome to Vegastrike
50
G: Particle,Particle,Ion
M: FriendOrFoe(1),Swarm(40),Swarm(40)
0
0
Fitchburg
Dis 48113.6719



gemini_sector/war:gemini_sector/war:unknown:25.40
8.55 hispidus: I hate you
15.00 nova: You are nice.
20.26 [Outgoing]: All is Fine
21.15 Ship destroyed: confed:firefly:Halo-0
21.15 Flightgroup Halo destroyed
24.43 firefly: *hit*
24.43 firefly: *hit*
24.44 firefly: *hit*

33



G: Laser,Laser,Meson,
Meson
M: Dumbfire(1)

0

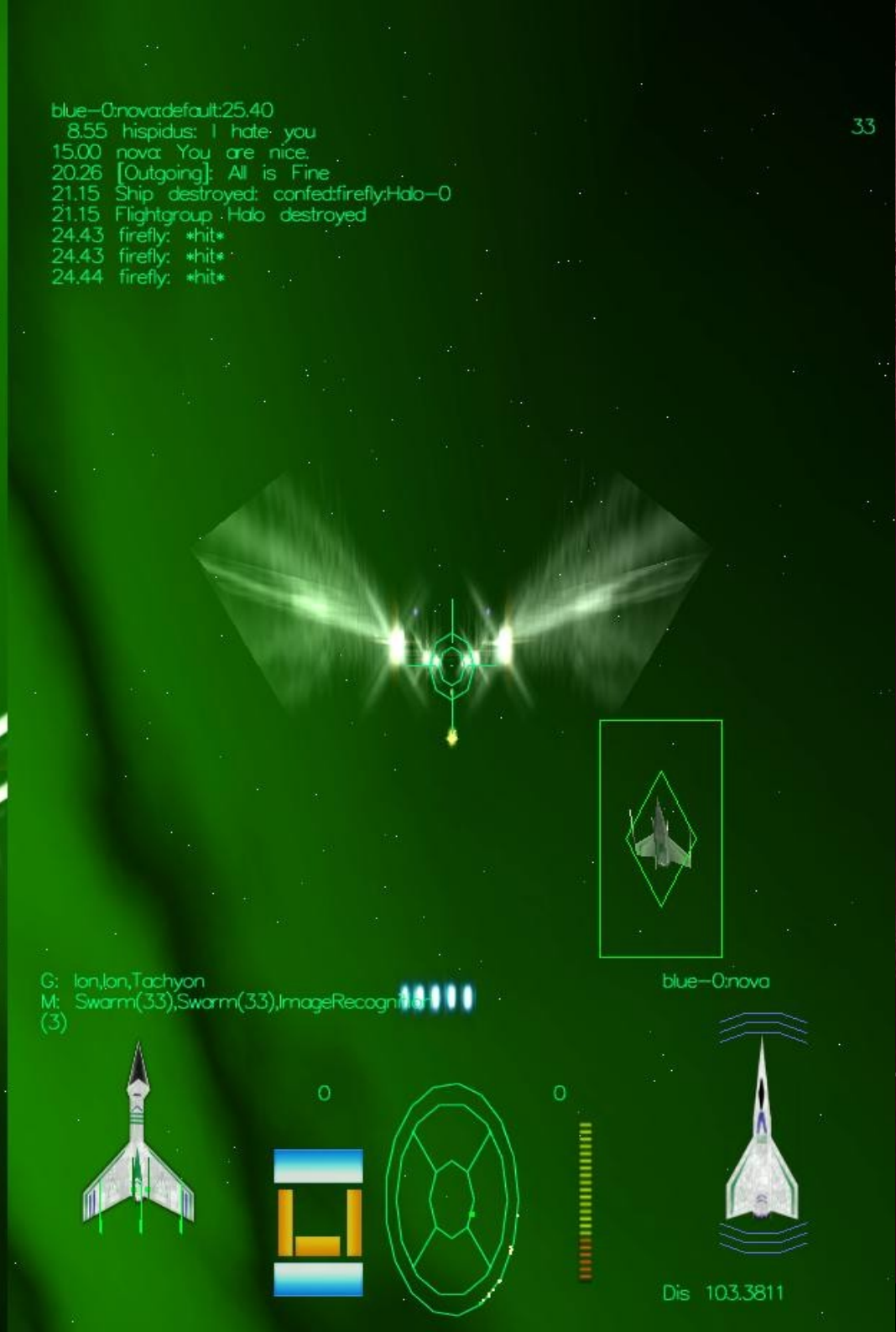
0

gemini_sector/war

Dis 11277.5029

blue-0:nova:default:25.40
8.55 hispidus: I hate you
15.00 nova: You are nice.
20.26 [Outgoing]: All is Fine
21.15 Ship destroyed: confed:firefly:Halo-0
21.15 Flightgroup Halo destroyed
24.43 firefly: *hit*
24.43 firefly: *hit*
24.44 firefly: *hit*

33



G: Ion,Ion,Tachyon
M: Swarm(33),Swarm(33),ImageRecogn
(3)

0

0

blue-0:nova

Dis 103.3811



Virtual Jay




0
7.81 / 24.00

X Ship Wars





	total	intimgs	isrefs	objs	ocsns
Mem:	5056952	906665	4137111	9792	1056
Ent:	223	123	182	19	22

Hail sent on channel 310.00

USS Stalion(#0P)
Shield Freq: 180.20 kHz
Weapon Freq: 160.00 kHz
Com Channel: 310.00 kHz



0.000 au/s

AM: 3999.67
HD: 19.23'
Sect: 0 0 0
X: 11.66
Y: -6.03

Eng:
(on)

WL: StarBase 902(#2H)
Stream Phazer: 100%
Photon Torp: 40(40)



Stream Phazer







0.3070au(9.60au)

StarBase 902(#2H)



0.0027 au

AM: 1000.00
HD: 0.00'
Sect: 0 0 0
X: 11.91
Y: -5.80

Int: 300 ms FPS: 10 SX: 241 RX: 1593 Lapse: 121

	total	intimgs	isrefs	objs	ocsns
Mem:	8274688	906665	7325311	14196	1056
Ent:	232	123	182	28	22

Hail sent on channel 310.00


Kertan(#0P)
Shield Freq: 180.20 kHz
Weapon Freq: 160.00 kHz
Com Channel: 310.00 kHz
Energy Conductor
0.1040au(10.40au)
Carrier 1 of 3(#1C)
0.036 au
0.000 au/s
AM: 6199.72
HD: 286.35'
Sect: 0 0 0
X: 5.61
Y: 37.78
Eng:
(on)
Cutting Beam: 100%
Pri Energy Weapon: 100%
Bolt Plasma: 100%
WL: Carrier 1 of 3(#1C)
Energy Conductor: 100%
AM: 30000.00
HD: 61.13'
Sect: 0 0 0
X: 5.22
Y: 37.56
Int: 125 ms FPS: 9 SX: 582 RX: 0 Lapse: 190
Hail sent on channel 310.00
Client RX SX Foreign Address State Player
std 64579 12940 207.168.6.251 ESTABLISHED Kertan(#0P)
Usage: 'netstat [player] [-l] [-c]'

IKS Dark Vixen(#0P)
Shield Freq: 180.20 kHz
Weapon Freq: 160.00 kHz
Com Channel: 310.00 kHz



0.000 au/s

AM: 999.80
HD: 71.76'
Sect: 0 0 0
X: -6.42
Y: -1.00

Eng:
(on)

WL: Earth(#3H)
Pulse Disruptor 2: 100%
Photon Torp: 30(30)
Stream Disruptor: 100%



Pulse Disruptor 2

2.5319au(9.60au)

Earth(#3H)



1.512 au

AM: 800.00
HD: 0.00'
Sect: 0 0 0
X: 12.16
Y: -4.51

Int: 550 ms FPS: 7 SX: 239 RX: 2257 Lapse: 150

External Lumination: Off
External Lumination: On
External Lumination: Off
Hail sent on channel 310.00




```
Events export log: /home/xsw/swserv/public_html/generic.events  420 bytes
Client RX           SX          Foreign Address                 State          Player
std    468662       85536       207.168.6.251                   ESTABLISHED    Guest 1(#1P)
Usage: 'netstat [player] [-l] [-c]'
```



Pulse Phazer 2: Power Yield: 20.00 Power Consumption: 6.00
Photon Torp: Power Yield: 20.00 Stock: 55(55)
Quantom Torp: Power Yield: 30.00 Stock: 35(35)
Stream Phazer: Unit Power Yield: 10.00 Unit Power Consumption: 2.50

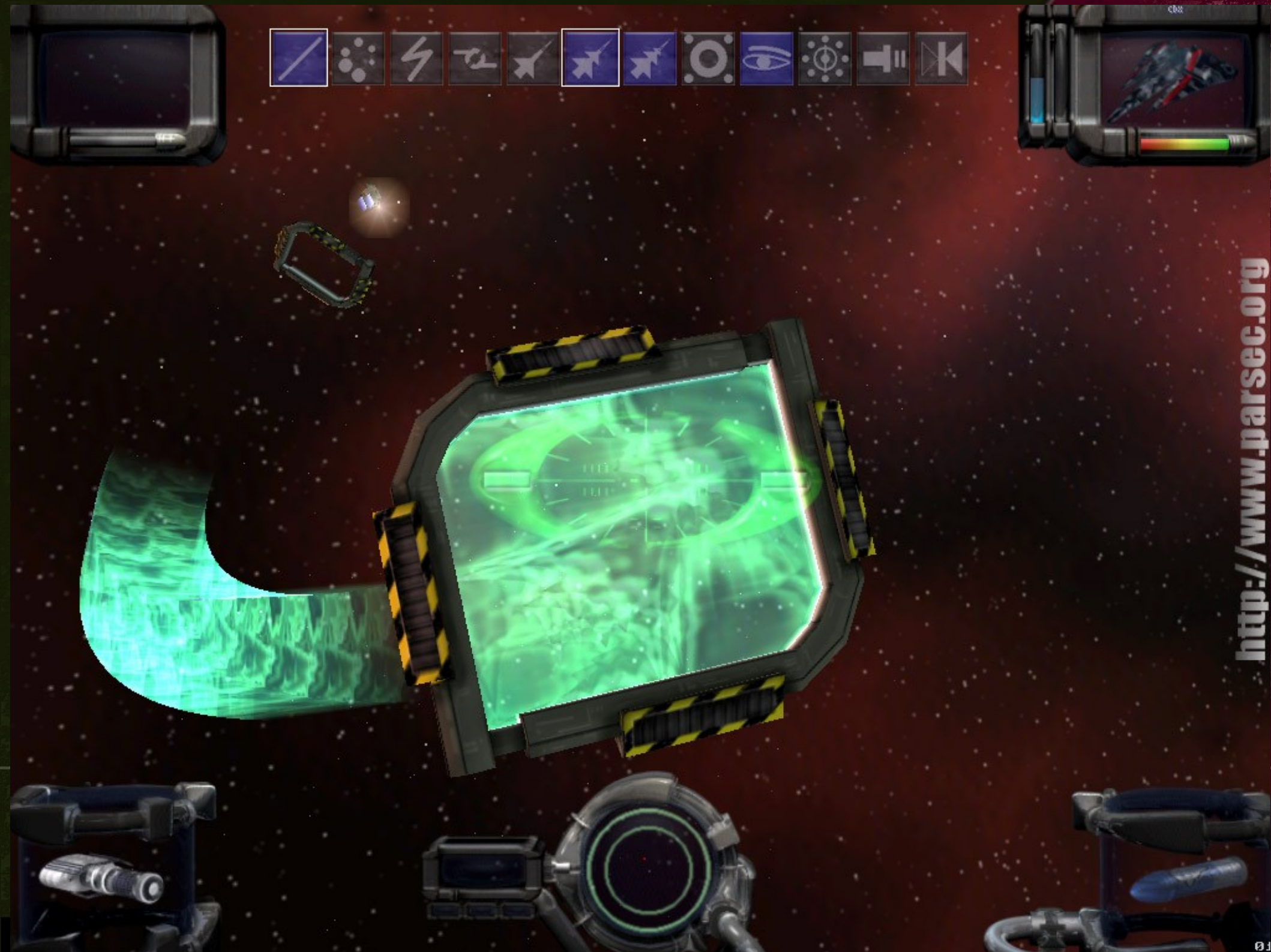
Parsec




MAVERICK
DISTANCE: 630
http://www.parsec.org
02


http://www.parsec.org


76
SIDWINGMAN
MAVERICK
http://www.parsec.org
SIDWINGMAN
DISTANCE: 195
OM-KOLTHOUM 02:07


http://www.parsec.org


SHIP STRUCK DOWN BY LIGHTNING
http://www.parsec.org

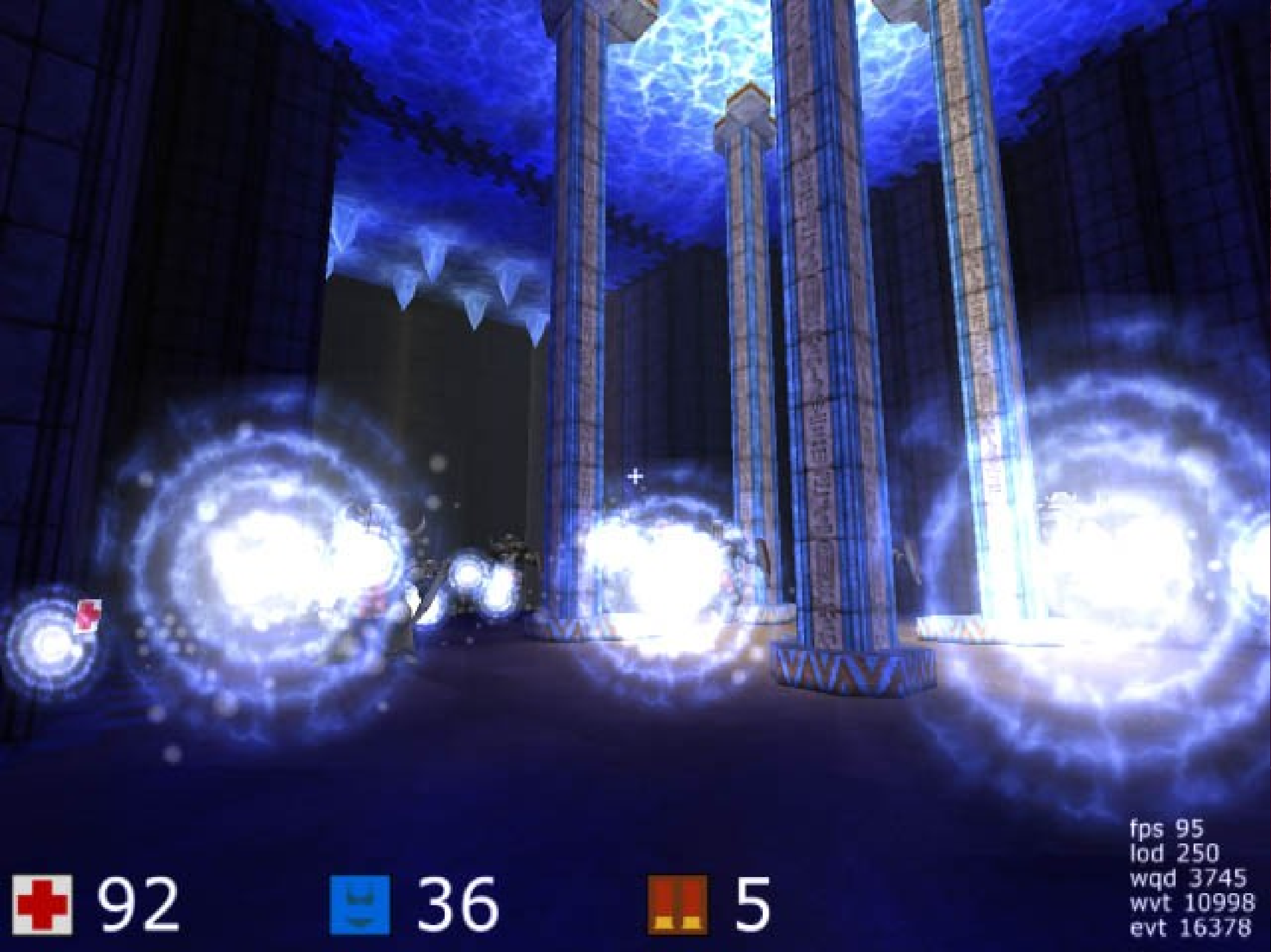

sid wingman
UNIMATRIX ZERO
PING: 29MS
GAMMA QUADRANT
PING: 5MS
http://www.parsec.org
OM-KOLTHOUM
01


http://www.parsec.org

Cube




quad damage is over
94
76
1
fps 87
lod 250
wqd 3192
wvt 10786
evt 13993


92
36
5
fps 95
lod 250
wqd 3745
wvt 10998
evt 16378


100
50
5
fps 109
lod 250
wqd 3850
wvt 13068
evt 1750

Torcs




3/10 - InfHist 4
Fuel: 7.7 l
Laps: 1 / 3
Best: 00:00 +00:00
Time: 01:12:87
<- InfHist 2 +00:00
-> InfHist 3 +00:00
FPS: 51.6
1: InfHist 1 00:00
2: InfHist 2 --:--
3: InfHist 4 --:--
4: InfHist 3 --:--
5: InfHist 9 --:--
6: InfHist 10 --:--
7: InfHist 7 --:--
8: InfHist 6 --:--
9: InfHist 5 --:--
10: InfHist 8 out
x1000 rpm
4
km/h
115
F
D



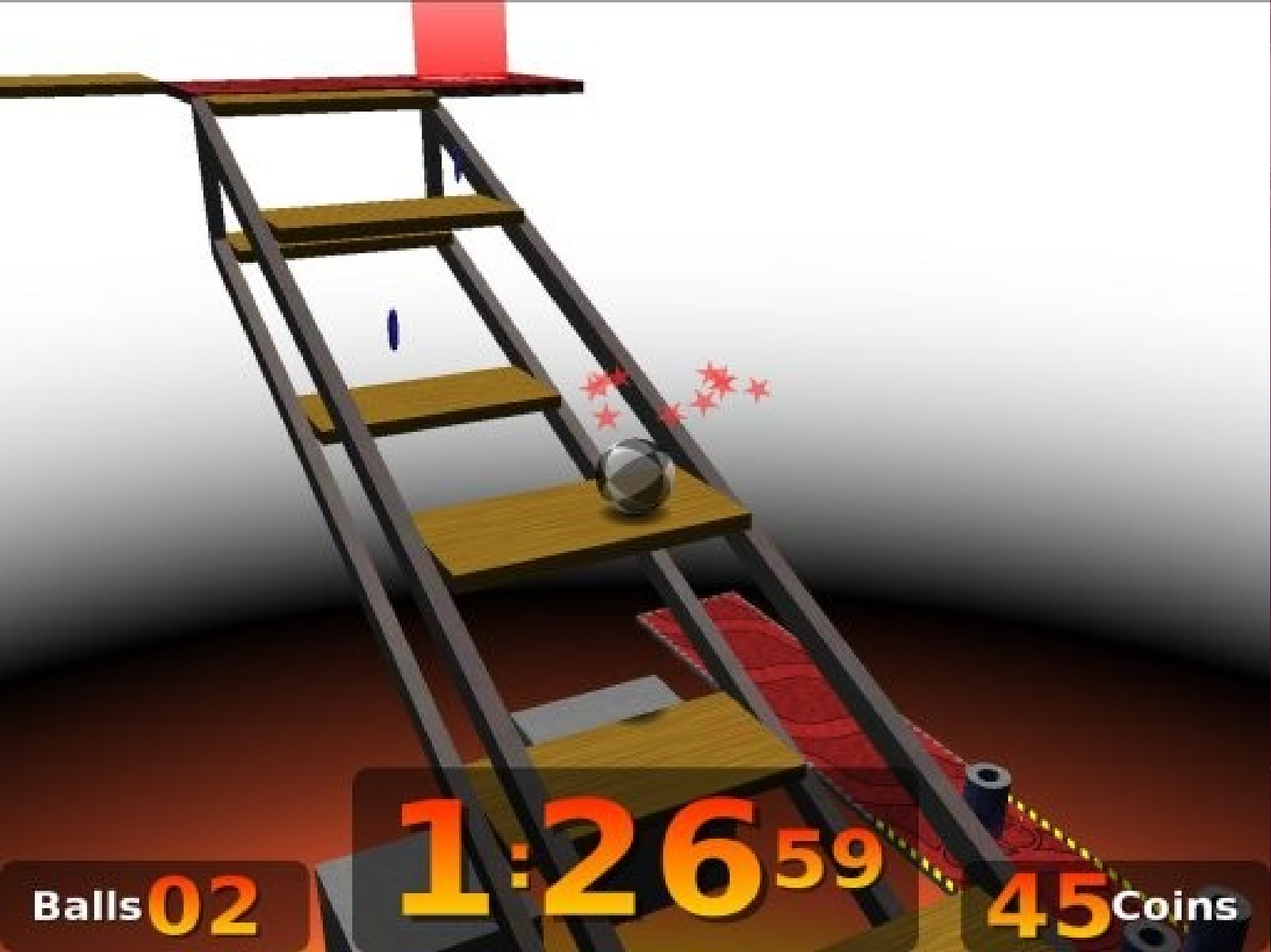
Neverball




0:59 60
Balls 02
19 Coins


1:05 25
Balls 02
11 Coins


Balls 02
0:27 08
03 Coins


Balls 02
1:26 59
45 Coins

Tuxracer




Time
01:25 98
x 000
174
km/h


Time
01:32 67
x 023
FINISH
36
km/h

UFO: Alien Invasion




OLIVIA GUADALUPE
1 2 3 4 5 6 7 8
8 7 6 5 4 3 2 1
1
HEALTH 17
MOVE 5
0
0


MAURO GUADALUPE
1 2 3 4 5 6 7 8
8 7 6 5 4 3 2 1
3
HEALTH 103
MOVE 19
0
0

1
2
3
4
5
6
7
8

KYOKO TANAKA

MAEMI TANAKA

JAN NELSON

SUN DONG

Primary

Secondary

Misc

Armor

SUN DONG

BACKPACK



RIGHT HAND



LEFT HAND



BELT

HOLSTER



SHOTGUN





PRIMARY:	SINGLE SHOT
SECONDARY:	DOUBLE SHOT
DAMAGE:	40 / 80
TIME UNITS:	8 / 12
RANGE:	9.4 / 9.4
SPREAD:	5.8 / 9.0
AMMO:	8


OLIVIA GUADALUPE
1 2 3 4 5 6 7 8
8 7 6 5 4 3 2 1
18
23
SUBMACHINE GUN
FULL AUTO (18) 22
18

Linux Oyunları (devam...)

- Quake serisi
- Simcity 3000
- Heavy Gear 2
- Heretic serisi
- Soldier of Fortune serisi
- Descent 3
- Deus Ex serisi
- Rune



Linux Oyunları (devam...)

- FAKK2
- Alpha Centauri
- Tribes 2
- Mind Rover
- Jagged Alliance 2
- Lgames (Lbreakout2, Ltris ...)
- Baldur's Gate
- KDE/GNOME games



Linux Oyunları (devam...)

- Unreal
- Unreal Tournament serisi (2004, 2003 ...)
- RTCW
- RTCW – ET
- ...



Farklı Platformlar İçin

- MAME (Xmame, Glmame, Advanced-mame)
- Amiga (UAE)
- Commodore (VICE)
- Sega Mastersystem/Genesis (Dega, Dgen)
- Nintendo/Nintendo64 (True Reality, gsnes9x)
- Playstation (pcsx)
- Gameboy (cingb, gnuboy)
- Atari2600 (Stella)



Wine/WineX

- Wine Is Not an Emulator
- Wine != WineX
- WineX özellikle DirectX uyumluluğuna odaklanmıştır (Direct3D, DirectInput ...)
- WineX Transgaming firmasının açık kaynak kodlu ve ticari bir ürünüdür



WineX ile Çalışan Birkaç Oyun

- Sims serisi
- Diablo serisi
- Max Payne serisi
- Half-Life ve modları
- SWAT 3
- Total Annihilation
- Dune 2000
- Nerf Arena



WineX ile Çalışan Birkaç Oyun

- Warcraft serisi
- Grim Fandango
- Hitman serisi
- Starcraft ve eklentileri
- Black & White
- 3D Mark 2000
- Mortal Kombat serisi
- Gunman Chronicles



WineX ile Çalışan Birkaç Oyun

- Carmageddon 3
- Daytona USA Deluxe
- Resident Evil serisi
- Red Alert serisi
- ...



Oyun Yazmak İstersem

- Programlama Dili (C, C++, Python, Java ...)
- Widget -> QT, GTK+
- 3D -> Mesa/OpenGL
- Ses -> Alsalib, OpenAL
- SDL (UT2004) image,net,ttf,sprite...
- Plib (Torcs, Flight Gear)
- Clanlib (Pingus, Clanbomber)
- PyGame (NeHe demoları)



Oyun Yazmak İstersem

- Linux Game development Center (lgdc.sunsite.dk)
- Çeşitli belgeler (Howto vs.)
- Açık Kaynak Kodlu oyunların kodu



Bazı Linux Oyun Firmaları

- idsoftware.com
- bioware.com
- epicgames.com
- hyperion-entertainment.com
- vvisions.com
- linuxgamepublishing.com
- transgaming.com



Faydalı Linkler

- happypenguin.org
- tuxgames.com
- linuxgames.com
- linuxgames.co.uk
- freshmeat.net
- sourceforge.net
- 3dgamer.com
- bluesnews.com





http://www.linux.org.tr
http://www.lkd.org.tr
http://seminer.linux.org.tr
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